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FE Practice Quiz M6

Question 1

Which of the following statements is **correct** regarding the residuals from spurious regression?

- The residuals have a unit root.
- The residuals are not serially correlated.
- The residuals can follow an MA(1) process.
- The value of DW test is around 0.
- ✗ No, the residuals are serially correlated.

Question 2

Which of the following statements is **correct** for Vector Autoregressive (VAR) model?

- All time series variables at time t will show on both sides of the equations in a VAR model.
- There are no constant terms or time trends in a VAR model.
- The error terms from different equations in a VAR model must be independent.
- VAR model can be used to model dynamic interactions of multiple time series.
- ✓ This is correct!

Question 3

Which of the following methods cannot test for stationarity of a time series?

- KPSS test
- Dickey-Fuller test
- Augmented Dickey-Fuller test

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✔ This is correct!

Question 4

Which of the following statements is **false** regarding the Johansen test?

- It is a likelihood-ratio test.
- The rank of the matrix that contains coefficients from a VAR model is used to decide cointegration.
- It can test cointegration for more than two time series.
- It includes trace test and maximum eigenvalue test.

✔ This is correct!

Question 5

If time series x_t and y_t are cointegrated, which of the following descriptions is correct?

- x_t and y_t are $I(0)$ and the residuals from the regression of x_t on y_t are $I(1)$
- x_t and y_t are $I(1)$ and the residuals from the regression of x_t on y_t are $I(1)$
- x_t and y_t are $I(0)$ and the residuals from the regression of x_t on y_t are $I(0)$
- x_t and y_t are $I(1)$ and the residuals from the regression of x_t on y_t are $I(0)$

✘ No. Both time series should be $I(1)$ and the residuals from the regression should be $I(0)$.

Question 6

Which of the following statements is the correct description of a 3-dimensional VAR(4) model?

QUESTIONS

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20

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equations, and 3 lags of all time series on the right side of the equations

- A VAR model with 3 time series, 4 equations, and 4 lags of all time series on the right side of the equations
- A VAR model with 4 time series, 3 equations, and 4 lags of all time series on the right side of the equations
- A VAR model with 3 time series, 3 equations, and 4 lags of all time series on the right side of the equations

✔ This is correct!

Question 7

Which of the following statements is **correct**?

- $I(1)$ is a time series integrated with order one.
- If $y_t \sim I(2)$, then $y_t - y_{t-1} = (1-B)y_t$ is a stationary time series.
- If $z_t \sim I(0)$, then z_t is a non-stationary time series.
- $I(1)$ is stationary.

✘ No, $I(1)$ has one unit root.

Question 8

Which test is used to check cointegration of more than two time series?

- Johansen test
- Engle Granger two-step test
- Dickey Fuller test
- Augmented Dickey Fuller test

✔ This is correct!

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Which of the following statements is **correct** for an error correction model?

- Error correction models only model long-term relationship of time series.
- Time series in an error correction model do not need to be cointegrated.
- The error correction coefficient decides the speed the time series will approach to stationary.
- Error correction models can model short-term and long-term dynamics of time series.

✔ This is correct!

Question 10

Which of the following is not a property of a VAR model?

- Simultaneous Model
- Stationary and ergodicity
- Seemingly Unrelated Regression
- Contemporaneous Correlation

✘ This is a property of VAR model. Seemingly unrelated regression means the error terms of the same time from different equations are correlated.

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hypothesis of the Dickey-Fuller test?

- The AR(1) model of the time series has a coefficient of less than 1.
- The time series is an explosive time series.
- The time series has a unit root.
- The time series does not have a unit root.

✔ This is correct!

Question 12

Which of the following statements is **correct** when nonstationary time series x_t, y_t and z_t are cointegrated?

- There exists a linear combination of x_t, y_t, and z_t, and the linear combination is stationary.
- We can use a VAR model to model x_t, y_t, and z_t.
- The multiplication of x_t, y_t, and z_t is stationary.
- The max of x_t, y_t, and z_t is stationary.

✔ This is correct!

Question 13

The Engle-Granger two-step procedure for testing cointegration is also called

- coefficient-based test for cointegration.
- residual-based test for cointegration.
- estimation-based test for cointegration.
- probability-based test for cointegration.

✔ This is correct!

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What tests do we usually do before running a VAR model?

- Augmented Dickey-Fuller test
- Shapiro-Wilk test
- Normal QQ test
- Ljung-Box test

✔ This is correct!

Question 15

Which of the following statements is **false** regarding unit root?

- Unit roots are roots of 1 from characteristic equations of an AR process.
- A time series with a unit root is stationary.
- Random walk process has a unit root.
- A time series can have more than 1 unit root.

✔ This is correct!

Question 16

Which of the following statements is **correct** regarding the integrated time series $I(5)$?

- The time series will be stationary after removing the trend.
- The level of time series is stationary.
- The first difference of the time series is stationary.
- The time series will be stationary after differencing it five times.

✔ This is correct!

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Which of the following statements is **correct** regarding spurious regression?

- Run two independent random walk time series on a regression and get a significant coefficient
- Run two independent random walk time series on a regression and get a 0 coefficient
- Run two independent non-stationary time series on a regression and get a 0 coefficient
- Run two independent stationary time series on a regression and get a significant coefficient

✗ No. A spurious regression issue requires running two independent nonstationary time series on a regression to get a significant coefficient.

Question 18

Which of the following statements is **correct** for ergodicity?

- An ergodic time series only needs data points to cover 10% of all possible values from its distribution.
- For an ergodic time series, the temporal sample mean is a consistent estimator of the mean of the process.
- An ergodic time series has a normal distribution.
- An ergodic time series does not require too many data points to achieve unbiased estimation.

✓ This is correct!

Question 19

Which of the following statements regarding stationarity is correct?

- ARMA model can directly model non-stationary time series.

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stationary by taking first difference.

- A time series is stationary if its mean and variance are independent of time.
- Difference-stationary time series cannot be made stationary.

✔ This is correct!

Question 20

Assume we have the following vector error correction model:

$$\Delta X_t = C + \Pi X_{t-1} + \Gamma \Delta X_{t-1} + \epsilon_t$$

Where X_t is a time series vector, ϵ_t is a white noise vector, and C is a deterministic vector.

Which term is the error correction term?

- $C + \Pi X_{t-1}$
- ΠX_{t-1}
- $\Pi X_{t-1} + \Gamma \Delta X_{t-1}$
- $\Gamma \Delta X_{t-1}$

✔ This is correct!

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